Algorithm: Comparison of Two Trees

ALGORITHM: MODIFIED COMPARISON OF TWO GENERAL TREES

1. Start.

2. Read the number of nodes, values, and child relationships for Tree 1.
3. Read the number of nodes, values, and child relationships for Tree 2.
4. Validate both trees: root has no parent, each non–root node has exactly one parent, all nodes are reachable, and no cy
5. If either tree is invalid, display an error and stop.
6. Calculate the node count and height of each tree.
7. Display both node counts and whether they match.
8. Display both heights and whether they match.
9. Recursively compare corresponding nodes from the roots.
10. If corresponding values differ, mark a value difference.
11. If corresponding child counts differ or a corresponding node is missing, mark a structural difference.
12. Save the first mismatch encountered in left–to–right order.
13. Report whether the trees are identical, differ only in structure, differ only in values, or differ in both.
14. Display the first mismatch, or “None” if no difference exists.
15. Free allocated memory and stop.

Note: Matching node counts and heights alone do not prove that two trees are identical.